Factorization of Polynomials Over the Boolean Semifield

IUP Scholars Forum

Micah Sherry
nbzx@iup.edu

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Background

The Boolean semifield, denoted as $\mathbb{B}$, is an algebraic structure with two operations: logical OR, analogous to addition, and logical AND, analogous to multiplication. $\mathbb{B}$ is the only finite semifield that is not a field. Our research explores the factorization of polynomials over the Boolean semifield, denoted as $\mathbb{B}[x]$.

A key challenge in $\mathbb{B}[x]$ is that factorization is not unique. In contrast the factorization of polynomials over fields is unique. Specifically, we investigate the factorization of a particular class of polynomials in $\mathbb{B}[x]$ namely:

$$f_n(x) = \sum_{i=0}^{n} x^i = x^n + x^{n-1} + \cdots + x + 1$$

Understanding the non-unique factorization of these polynomials is a central focus of our study. This research contributes to a deeper understanding of algebraic structures beyond traditional fields and rings.

Definitions

- A field is an algebraic structure Like the real numbers. Where operations like addition, multiplication, subtraction and division can be performed.
- A semifield is an algebraic structure like a field, except subtraction can not be performed.
- Let $\mathbb{B} = \{0, 1\}$ be the Boolean semifield equipped with logical OR as addition and Logical AND as Multiplication.
  - So, $1 + 1 = 1$.
- Let $\mathbb{B}[x]$ be the set of all polynomials in x over $\mathbb{B}$.
- Let $f_n(x) \in \mathbb{B}[x]$, be defined as $f_n(x) = x^n + x^{n-1} + \cdots + x + 1$.
- We say a polynomial $f(x)$ is factorable if $f(x) = g(x)h(x)$ for some $g(x), h(x) \neq f(x)$

Example Multiplications

$$\begin{aligned}(x^2+1)(x^2+1) &= x^2(x^2+1)+(x^2+1)\\ &= x^4+x^2+1\end{aligned}$$

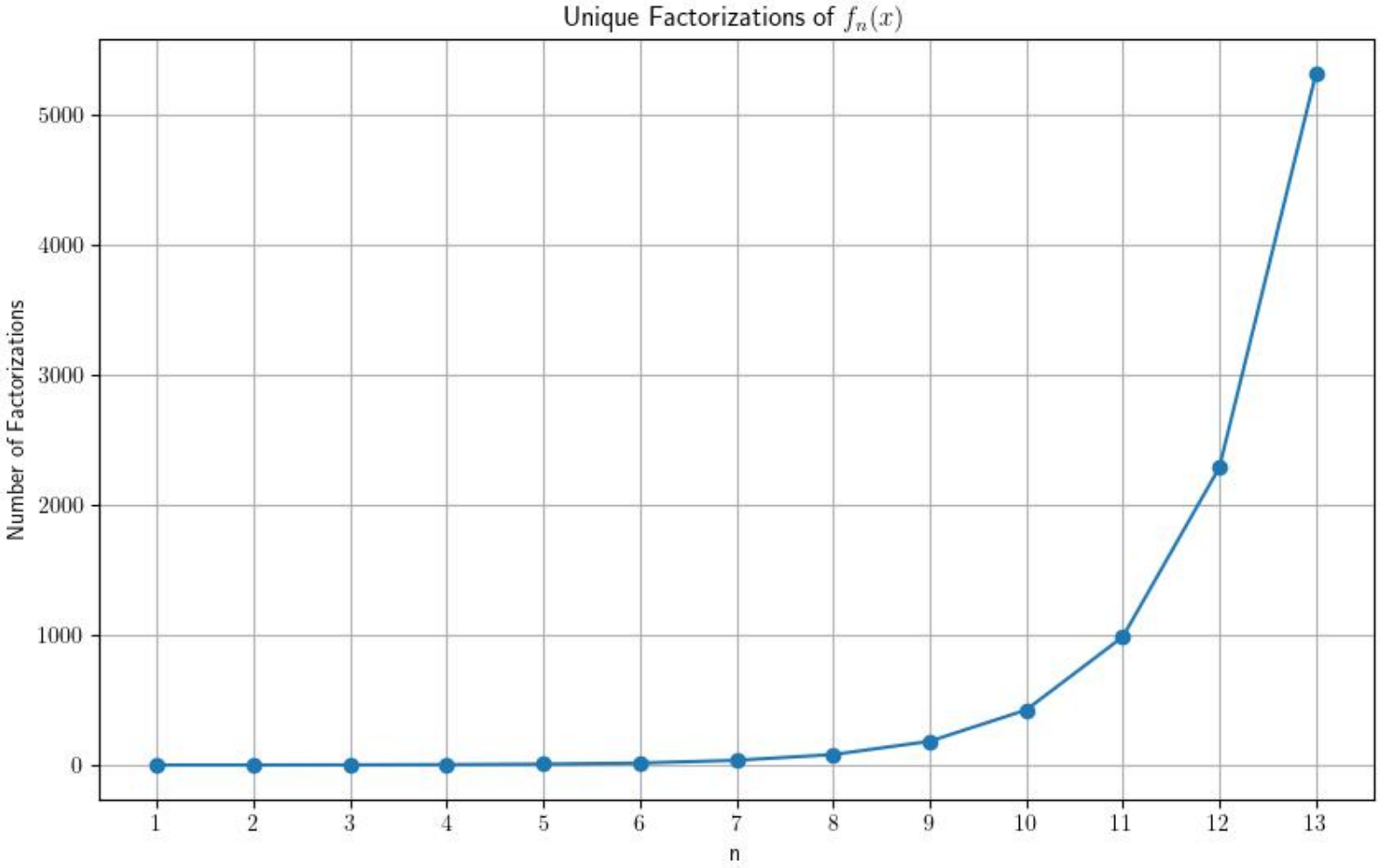
$$\begin{aligned}(x^3+1)(x^2+1) &= x^3(x^2+1)+(x^2+1)\\ &= x^5+x^3+x^2+1\end{aligned}$$

$$\begin{aligned}(x^4+1)(x^3+x+1) &= x^4(x^3+x+1)+(x^3+x+1)\\ &= x^7+x^5+x^4+x^3+x+1\end{aligned}$$

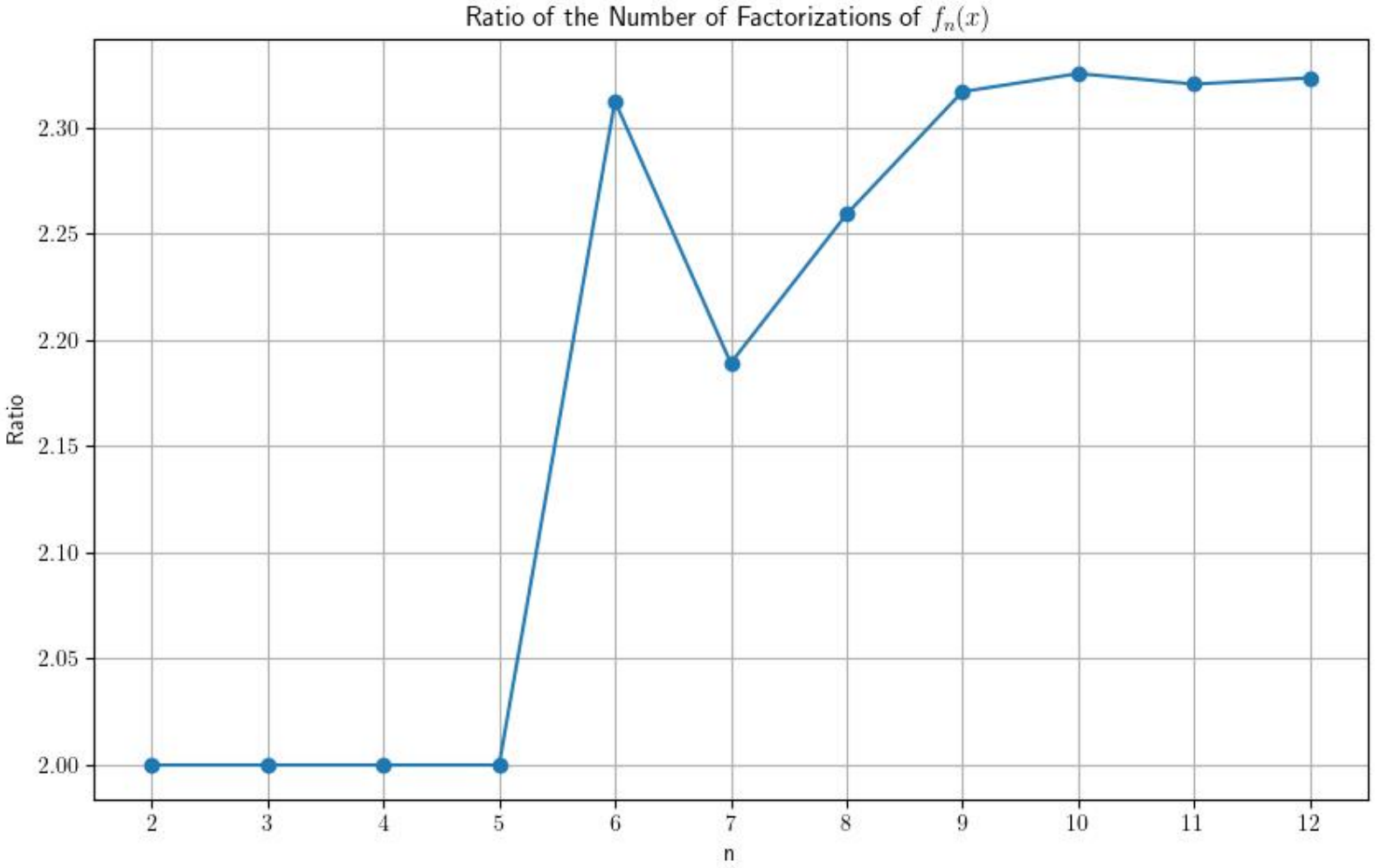
Examples of Possible Factorization

n	$f_n(x)$	Factorizations
2	$x^2 + x + 1$	$(x+1)^2$
3	$x^3 + x^2 + x + 1$	$(x+1)^3$ $(x+1)(x^2+1)$
4	$x^4 + x^3 + x^2 + x + 1$	$(x+1)^4$ $(x+1)^2(x^2+1)$ $(x+1)(x^3+x^2+1)$ $(x+1)(x^3+x+1)$
5	$x^5 + x^4 + x^3 + x^2 + x + 1$	$(x+1)^5$ $(x+1)^3(x^2+1)$ $(x+1)^2(x^3+x^2+1)$ $(x+1)^2(x^3+x+1)$ $(x+1)^2(x^3+1)$ $(x+1)(x^4+x^3+x^2+1)$ $(x+1)(x^4+x^2+x+1)$ $(x+1)(x^2+1)^2$

Exponential Growth of Factors



Growth Rate of Factors



Necessary but Not Sufficient Condition for Factorization

Theorem 1

Let $f(x) = \sum_{i=0}^{n} f_i x^i$ *with* $f_0 = 1$. *If* $f(x)$ *is factorable, then there exist* j, k *with* $1 \leq j, k \leq n-1$ *and* $j + k = n$ *such that* $f_j = f_k = 1$.

Proof.

Assume $f(x) = g(x)h(x)$ with $1 \leq \deg(g), \deg(h) \leq n-1$. Let $\deg(g) = j$, $\deg(h) = k$. Then $g(x) = x^j + g'(x) + 1, \quad h(x) = x^k + h'(x) + 1$.

$$\begin{aligned} f(x) &= g(x)h(x) \\ &= (x^j + g'(x) + 1)(x^k + h'(x) + 1) \\ &= x^{j+k} + x^j + x^k + \big(x^j h'(x) + x^k g'(x) + g'(x)h'(x) + g'(x) + h'(x)\big) + 1. \end{aligned}$$

Thus $j + k = n$ and $f_j = f_k = 1$ □

Remarks.

$f(x) = x^6 + x^4 + x^2 + x + 1$ satisfies the condition but is not factorable.

$f(x) = x^6 + x^5 + x^2 + 1$ does not satisfy the condition and is not factorable.

Factorizations Of The Form $(x+1)h(x) = f_n(x)$

Theorem 2

$(x+1)h(x) = f_n(x)$ *if and only if* $h(x) = \sum_{i=0}^{n-1} h_i x^i$ *With* $h_{n-1} = h_0 = 1$ *and* $h_i + h_{i-1} = 1$ *for* $0 < i < n-1$.

Proof.

($\rightarrow$): Assume that $(x+1)h(x) = f_n(x)$ and let $h(x) = \sum_{i=0}^{n-1} h_i x^i$. (Notice $h_{n-1} = h_0 = 1$ otherwise it would contradict the assumption). Now assume for the sake of contradiction that there exists $k < n-1$ such that $h_k = h_{k-1} = 0$. Consider the following:

$$(x+1)h(x) = x\sum_{i=0}^{n-1} h_i x^i + \sum_{i=0}^{n-1} h_i x^i.$$

Notice the k term will have coefficient $h_k + h_{k-1} = 0$, contradicting the original assumption.
($\leftarrow$): Assume $h(x) = \sum_{i=0}^{n-1} a_i x^i$ with $h_{n-1} = h_0 = 1$ and for all h_i, $h_i + h_{i-1} = 1$. Consider the following:

$$(x+1)h(x) = \sum_{i=0}^{n-1} a_i x^{i+1} + \sum_{i=0}^{n-1} a_i x^i = a_{n-1}x^n + \sum_{i=1}^{n-1}(a_{i-1} + a_i)x^i + a_0 = f_n(x).$$

Therefore, the theorem holds. $\square$

Factorizations Of The Form $(x+1)h(x) = f_n(x)(Cont.)$

Theorem 3

Let $G_{n\,1} = \{h(x) | (x+1)h(x) = f_n(x)\}$ *Let* F_n *be the nth Fibonacci number.* $|G_{n\,1}| = F_n$ *for* $n \geq 1$

Proof (by strong induction).

$G_{1\,1} = \{1\}, \quad |G_{1\,1}| = 1 = F_1 \quad \& \quad G_{2\,1} = \{x+1\}, \quad |G_{2\,1}| = 1 = F_2$
Therefore the base cases hold. Assume that $|G_{j\,1}| = F_j$ for all $j \in \{1, 2, \ldots, k\}$ for some k.
Let $h(x) \in G_{(k+1)\,1}$. Notice $\deg(h(x)) = k$ and $h_k = 1$ consider the possible values for h_{k-1}.

1. Case 1. $h_{k-1} = 1$. In which case $h(x) = x^k + h'(x)$ Where $h'(x) \in G_{k\,1}$.
2. Case 2. $h_{k-1} = 0$, So $h_{k-2} = 1$. In which case $h(x) = x^k + 0x^{k-1} + h''(x)$ Where $h''(x) \in G_{(k-1)\,1}$.

So, $|G_{(k+1)\,1}| = |G_{k\,1}| + |G_{(k-1)\,1}| = F_k + F_{k-1} = F_{k+1}$. Therefore, by the Principle of Mathematical Induction $|G_{n\,1}| = F_n$ for all $n \geq 1$. □

Example $(x+1)h(x) = f_5(x)$

Consider the polynomials $h(x)$ such that $(x+1)h(x) = f_5(x)$ (i.e. $G_{5\,1}$).

$h(x) \in G_{5\,1}$
$x^4 + x^3 + x^2 + x + 1$
$x^4 + x^3 + x^2 \qquad + 1$
$x^4 + x^3 \qquad + x + 1$
$x^4 \qquad + x^2 + x + 1$
$x^4 \qquad + x^2 \qquad + 1$

Notice there are $5 = F_5$ polynomials as expected.

Motivating example

Consider the Factorizations of the form

$$(x^2 + 1)g(x) = f_7(x)$$

the polynomials $g(x)$ are listed below.

$g(x)$
$x^5 + x^4 + x^3 + x^2 + x + 1$
$x^5 + x^4 + x^3 \qquad + x + 1$
$x^5 + x^4 \qquad + x^2 + x + 1$
$x^5 + x^4 \qquad\qquad + x + 1$

Is there a general formula for the number of factorizations like $(x^k + 1)g(x) = f_n(x)$?

Factorizations Of The Form $(x^k + 1)g(x) = f_n(x)$

Theorem 4

Let $G_{n\,k} = \{g(x) | (x^k + 1)g(x) = f_n(x)\}$, *let* $m = n - 3k + 1$, *let* $q = \lfloor \frac{m}{k} \rfloor$, *and let* $r = m \bmod k$.

$$|G_{n\,k}| = (F_{q+3})^r \times (F_{q+2})^{k-r}$$

Sketch.

Consider $g(x) = \sum_{i=0}^{n-k} g_i x^i \in G_{nk}$.

$$\begin{aligned}(x^k + 1)g(x) &= \sum_{i=0}^{n-k} g_i x^{i+k} + \sum_{i=0}^{n-k} g_i x^i \\ &= \sum_{i=k}^{n} g_{i-k} x^i + \sum_{i=0}^{n-k} g_i x^i \\ &= \sum_{i=n-k+1}^{n} g_i x^i + \sum_{i=k}^{n-k} (g_k + g_{i-k}) x^i + \sum_{i=0}^{k-1} g_i x^i = f_n(x)\end{aligned}$$

Notice this implies $g_i = 1$ for $i \leq k - 1$ or $i \geq n - 2k + 1$ and $g_i + g_{i-k} = 1$ for $k \leq i \leq n - 2k$

Factorizations Of The Form $(x^k + 1)g(x) = f_n(x)$ (Cont.)

Sketch.

Construct a partition the coefficients g_i where $k \leq i \leq n - 2k$. Let $g_i \in P_j$ if $i \equiv j \mod k$.

P_0	$g_k,\ g_{2k},\ \cdots$	$i \equiv 0 \mod k$
P_1	$g_{k+1},\ g_{2k+1}, \cdots$	$i \equiv 1 \mod k$
$\vdots$	$\vdots$	$\vdots$
P_{k-1}	$g_{2k-1},\ g_{3k-1}, \cdots$	$i \equiv k-1 \mod k$

Let $m = n - 3k + 1$, $q = \lfloor \frac{m}{k} \rfloor$, and $r = m \mod k$. The length of r of these partitions is $q + 1$ and $k - r$ have length q. It can be shown that the number of options (where $g_i + g_{i-k} = 1$) for P_j of length q is F_{q+2}. So, The number of valid polynomials is $(F_{q+3})^r \times (F_{q+2})^{k-r}$ $\square$

Example $n = 20$ and $k = 4$

if $(x^4 + 1)g(x) = f_{20}(x)$ then

$$g(x) = \sum_{i=0}^{16} g_i x^i$$

with $g_i = 1$ for $i \leq 3$ or $i \geq 13$ and $g_i + g_{i-4} = 1$ for $2 \leq i \leq 12$

P_i	$g_i \in P_i$	#
P_0	g_4, g_8, g_{12}	5
P_1	g_5, g_9	3
P_2	g_6, g_{10}	3
P_3	g_7, g_{11}	3

Notice $m = n - 2k + 1 = 9$, $q = \lfloor \frac{m}{k} \rfloor = 2$, and $r = m \mod k = 1$. So, The Number of valid g(x) is $(F_5) \times (F_4)^3 = 5 \times 3^3 = 135$

Theorem 5

Let $P(g(x))$ be the number of unique factorizations of $g(x)$.

$$F_{n+2} - n \leq P(f_n(x))$$

Proof (by induction).

Consider the case when $n = 1$. Notice $F_3 - 1 = 1$ and $f_1(x) = x + 1$ only has one factorization. Thus the base case holds. Assume that there exist a k such that $F_{k+2} - k \leq P(f_k(x))$. Consider the case for f_{k+1}. From Theorem 3 it follows that,

$$F_{k+1} \leq P(f_{k+1}(x)).$$

Notice that $(x + 1)f_k(x)$ is a factorization for f_{k+1} so,

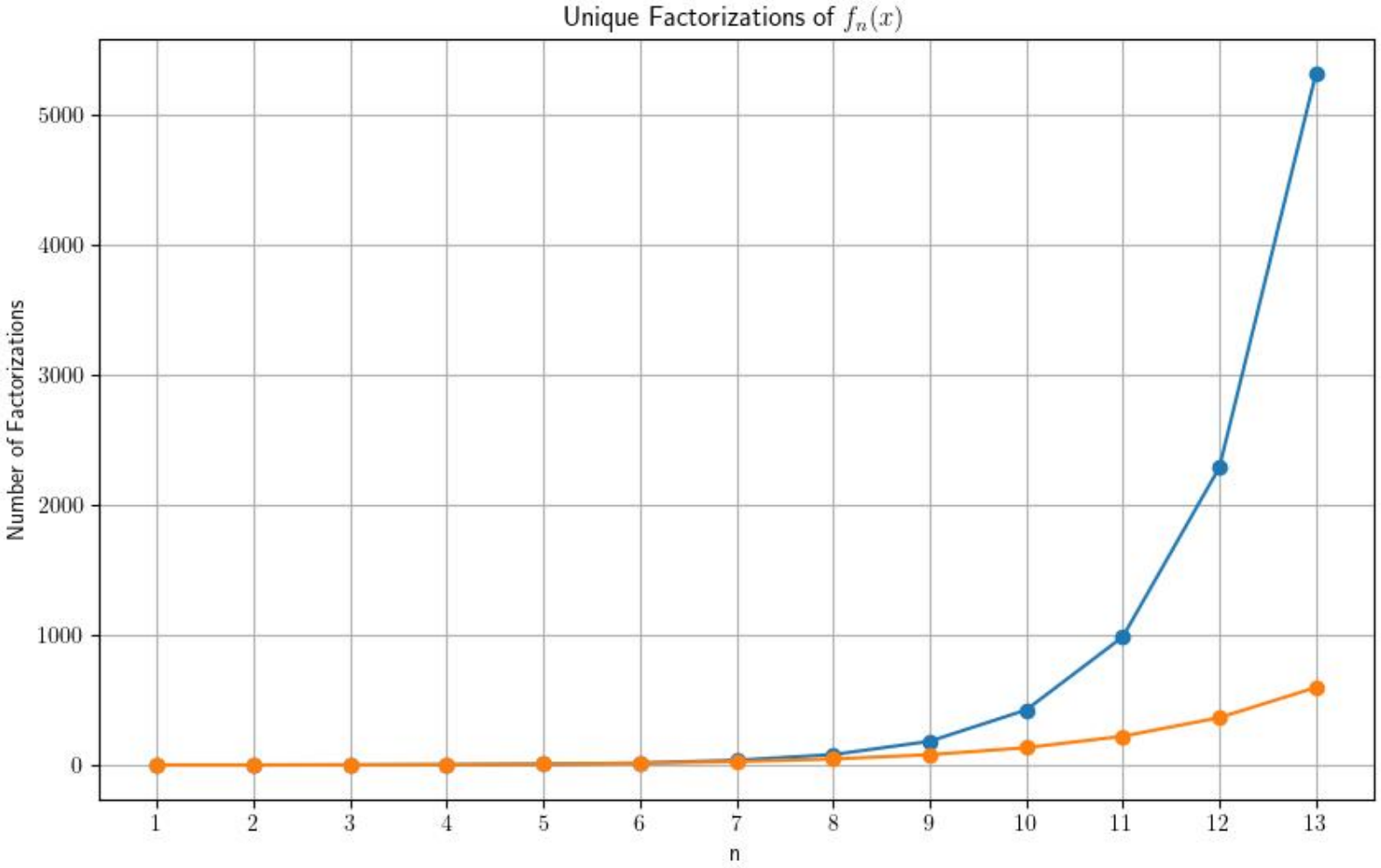
$$P(f_k(x)) + F_{k+1} - 1 \leq P(f_{k+1}(x)).$$

By the induction hypothesis

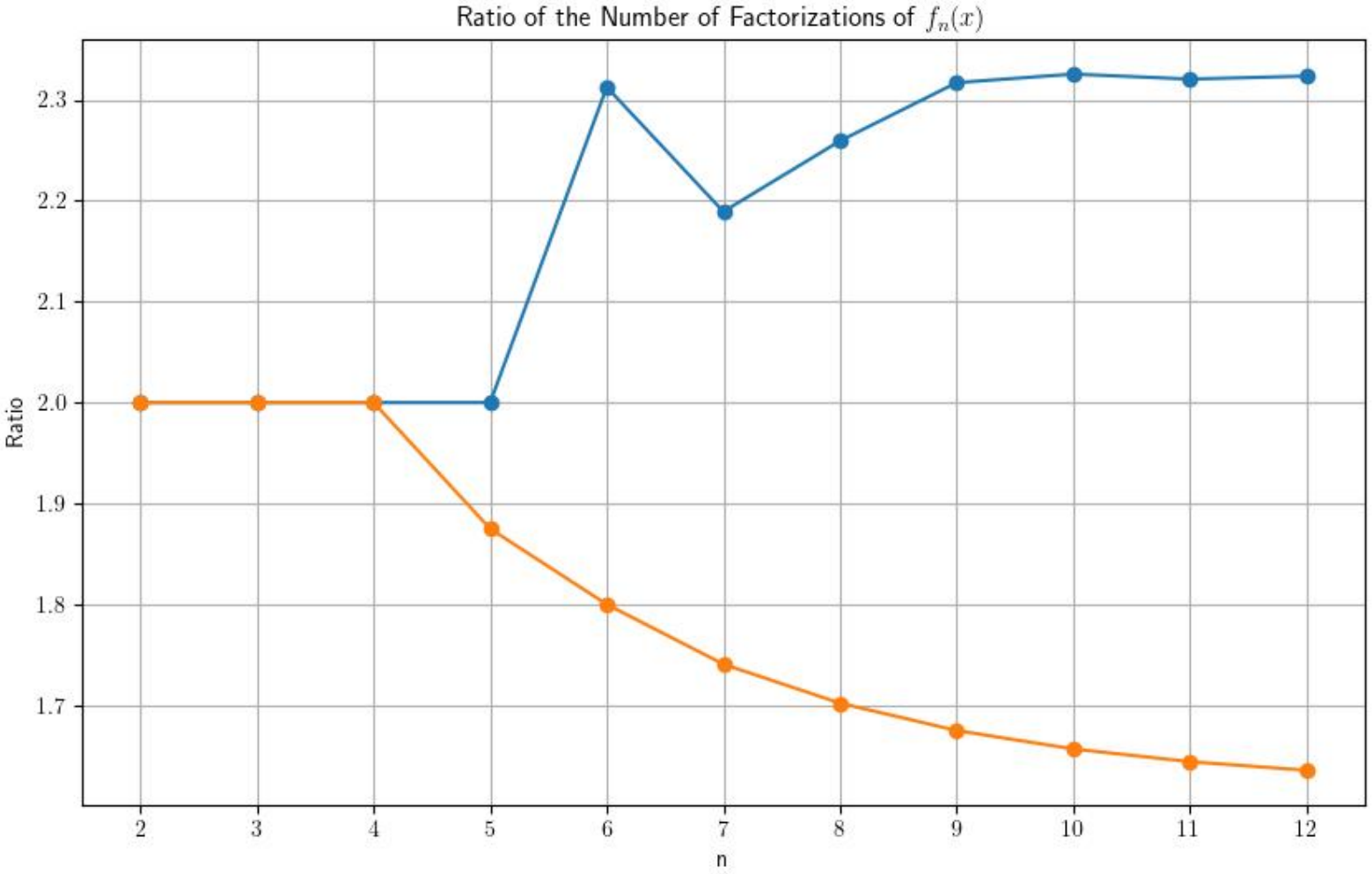
$$F_{k+2} + F_{k+1} - (k + 1) \leq P(f_k(x)) + F_{k+1} - 1 \leq P(f_{k+1}(x)).$$

Thus by the principle of mathematical induction $F_{n+2} - n \leq P(f_n(x))$. $\square$

Exponential Growth of Factors



Growth Rate of Factors



Sources

[1] Francisco Alarcon and Daniel D Anderson.
Commutative semirings and their lattices of ideals.
Houston J. Math, 20(4):571–590, 1994.

[2] David Applegate, Marc LeBrun, and Neil JA Sloane.
Dismal arithmetic.
arXiv preprint arXiv:1107.1130, 2011.

[3] Andrea Frosini, S Rinaldi, and Pian dei Mantellini.
On the sequence a079500 and its combinatorial interpretations.
J. Integer Seq, 9, 2006.

[4] Gal Gross.
Maximally additively reducible subsets of the integers.
University of Toronto (Canada), 2019.

[5] Ki Hang Kim and Fred W Roush.
Factorization of polynomials in one variable over the tropical semiring.
arXiv preprint math/0501167, 2005.

[6] OEIS Foundation Inc.
A067139: Irreducible elements in or-numbral arithmetic.
The On-Line Encyclopedia of Integer Sequences, 2002.
Available at `https://oeis.org/A067139`.

[7] OEIS Foundation Inc.
A169912: Number of irreducible boolean polynomials of degree n.
The On-Line Encyclopedia of Integer Sequences, 2010.
Available at `https://oeis.org/A169912`.

[8] OEIS Foundation Inc.
A171000: Irreducible boolean polynomials written as binary vectors.
The On-Line Encyclopedia of Integer Sequences, 2010.
Available at `https://oeis.org/A171000`.